

SMITHERS, BC, Canada

WMO: 719500

Lat: 54.8242N

Lon: 127.1894W

Elev: 523

StdP: 95.20

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-21.5	-29.4	0.3	-24.4	-25.6	0.4	-20.6	8.2	-1.8	7.3	-2.9	1.1	150	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	27.8	16.4	25.3	15.4	23.1	14.4	17.3	25.1	16.2	23.5	15.3	21.5	2.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.0	19.1	13.5	10.3	18.0	12.6	9.7	17.2	50.5	25.1	47.3	23.6	44.4	21.8	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.6	5.1	DB	-28.6	31.2	4.1	2.3	-31.5	32.8	-33.9	34.2	-36.2	35.5	-39.1	37.2	
			WB	-28.8	19.5	4.0	1.6	-31.7	20.6	-34.0	21.6	-36.3	22.4	-39.2	23.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-6.7	-4.8	-0.4	4.8	9.7	13.1	15.3	14.7	10.2	4.4	-1.9	-6.8
	DBStd	9.11	7.23	5.90	4.89	2.98	3.42	3.10	2.93	2.96	3.15	3.10	4.73	6.31
	HDD10.0	2551	516	415	321	159	48	6	1	1	34	174	356	520
	HDD18.3	5119	774	648	579	407	268	161	103	119	243	431	606	778
	CDD10.0	492	0	0	0	2	39	97	164	147	41	2	0	0
	CDD18.3	17	0	0	0	0	0	3	8	6	0	0	0	0
	CDH23.3	453	0	0	0	1	15	83	185	159	11	0	0	0
	CDH26.7	112	0	0	0	0	1	18	54	38	1	0	0	0
Wind	WSAvg	1.6	1.7	1.5	1.9	1.9	1.8	1.7	1.5	1.4	1.3	1.6	1.7	1.7
Precipitation	PrecAvg	539	62	32	25	20	36	44	47	45	55	67	57	50
	PrecMax	741	125	71	68	42	75	89	120	96	87	119	119	110
	PrecMin	371	14	4	7	4	8	10	11	4	7	28	7	8
	PrecStd	71	34	14	14	10	17	24	25	24	20	23	26	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.5	12.5	20.4	25.3	29.2	31.6	30.1	24.9	15.6	8.8	5.7
		MCWB	3.5	3.2	6.2	10.3	13.2	16.4	17.2	16.9	15.2	11.0	5.9	3.1
	2%	DB	4.3	4.6	9.2	15.4	22.5	25.7	27.8	27.5	21.1	12.8	6.3	3.4
		MCWB	2.5	2.0	4.3	7.4	12.1	15.2	16.4	16.6	13.8	9.0	4.4	1.8
	5%	DB	2.7	3.3	7.4	13.0	20.1	23.0	25.1	24.8	18.4	11.0	4.8	1.8
		MCWB	1.5	1.2	3.3	6.3	11.1	14.0	15.6	15.6	12.7	7.9	3.4	0.8
	10%	DB	1.2	2.0	5.8	11.0	17.5	20.4	22.5	22.1	16.1	9.3	3.3	0.5
		MCWB	0.5	0.5	2.3	5.2	10.0	13.0	14.9	14.5	11.5	6.7	2.2	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	3.7	6.7	10.9	14.4	17.4	19.1	18.8	16.1	12.1	6.4	3.5
		MCDB	5.8	5.7	11.7	18.9	22.8	26.3	28.0	26.8	22.8	14.6	7.9	5.4
	2%	WB	2.7	2.4	4.8	8.4	12.7	15.9	17.5	17.2	14.7	9.7	4.7	2.0
		MCDB	3.9	3.9	8.3	13.9	20.1	24.4	25.0	25.1	19.7	12.1	5.9	3.1
	5%	WB	1.6	1.5	3.7	6.9	11.6	14.6	16.4	16.1	13.3	8.2	3.6	0.9
		MCDB	2.6	2.9	6.9	11.6	18.3	21.3	23.2	23.5	17.4	10.3	4.6	1.7
	10%	WB	0.5	0.6	2.6	5.8	10.6	13.5	15.4	15.0	12.1	7.0	2.4	-0.1
		MCDB	1.2	1.8	5.3	10.0	16.5	19.3	21.4	20.9	15.3	8.9	3.2	0.5
Mean Daily Temperature Range	5% DB	MDBR	6.1	8.2	9.1	11.0	12.3	11.8	12.0	12.5	10.9	7.9	5.2	5.8
		MCDBR	6.4	7.9	10.9	15.2	17.5	17.3	17.7	18.1	15.5	10.7	6.7	6.3
		MCWBR	5.0	5.7	6.8	8.5	8.3	7.9	7.6	8.1	8.5	7.1	5.1	5.3
	5% WB	MCDBR	6.1	7.1	10.4	13.0	14.9	15.3	15.2	16.0	13.5	9.5	6.3	6.1
		MCWBR	4.9	5.3	6.7	7.8	7.4	7.2	7.1	7.6	7.7	6.6	4.9	5.2
Clear-Sky Solar Irradiance	taub	0.268	0.277	0.304	0.356	0.374	0.378	0.363	0.359	0.324	0.311	0.277	0.247	
	taud	2.199	2.273	2.217	2.133	2.258	2.350	2.445	2.463	2.553	2.507	2.272	2.201	
	Ebn at Noon	668	814	868	862	868	868	876	859	851	776	654	612	
	Edh at Noon	59	79	105	133	125	116	103	96	77	64	55	47	
All-Sky Solar Radiation	RadAvg	0.59	1.45	2.61	4.21	5.03	5.52	5.32	4.52	2.91	1.52	0.68	0.45	
	RadStd	0.07	0.19	0.26	0.24	0.43	0.53	0.54	0.23	0.30	0.20	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+59	N/A

Nomenclature: See separate page